

Hannah Laurie Lawson

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Education

PhD	Information Science, University of Colorado Boulder	Current
MS	Information Science, University of Colorado Boulder	May 2024
BS	Computer Science & Physics, Washington and Lee University	May 2021
	Minor Middle East and South-East Asia Studies with Arabic Language Focus	

Research Experience

Wikimedia Foundation, Graduate Researcher Aug. 2023 – April 2024

Prof. Brian C. Keegan

Network Perspectives on Multilingual Collective Memory Processes on Wikipedia

Grant Information:

https://meta.wikimedia.org/wiki/Grants:Programs/Wikimedia_Research_Fund/Network_perspectives_on_collective_memory_processes_across_the_Arabic_and_English_Wikipedias

- Developed and deployed a streamlined multilingual data preprocessing pipeline
- Spearheading groundbreaking integration of in low-resource and high-resource Natural Language Processing tools and Network Analysis tools into multilingual research pipelines
- Demonstrating a strong commitment to inclusion by actively integrating diverse perspectives into NLP model training, resulting in models that exhibit greater cultural sensitivity and fairness.

Preprint: <https://arxiv.org/pdf/2404.10706?>

UMD Applied Research Lab for Intelligence & Security, June. 2023 – Aug. 2023
Research for Intelligence & Security Challenges (RISC)
Summer Internship for Hard Security Problems

- Generated a scalable synthetic pattern of life data pipeline through Hidden Markov Models through diverse, open source, data sources
- Working seamlessly with cross-functional teams of diverse backgrounds and disciplines.
- Initiated and led a research project focused on data generation, contributing to a published poster at a conference of many high-level government officials

Los Alamos National Laboratory, Graduate Research Assistant May. 2022 – Dec. 2022

- Application of NLP tools to news media sources to extract and characterize narratives
- Internal data exploration and analysis leading to group wide presentations with laboratory wide repercussions

University of Colorado and NSF, Graduate Research Assistant Jan. 2022 – April 2022

Prof. Brian C. Keegan, Prof. Estelle Smith

Sense of Virtual Community on Reddit

- Supported an NSF-sponsored project utilizing NLP and network analysis to study genealogical relationships and behavioral sequences among online communities.

Sociology, Washington and Lee University, Undergraduate Researcher Jan. 2021 – April 2021

Prof. Jon Eastwood

Analyzing Trust within Middle Eastern Communities

- Conducted a literature review of sociology concepts and methods
- Determined notable information to perform regression analysis on the qualitative data in the World Value Survey.

Socio-Physics, Washington and Lee University, Undergraduate Researcher

Aug. 2020 – Dec. 2020

Prof. Irina Mazilu

What makes literature Internationally Famous?

- Creating a text database of internationally known fiction literature published between the 17th and 18th century
- Utilizing text analysis to draw correlations about common words, phrasing, and characteristics to identify trends that could correlate with international success

Computer Science, Washington and Lee University,
Undergraduate Researcher

March. 2020 – Aug. 2020

Dr. Cody Watson

Modeling Coronavirus in a Small University in Vensim

- Used virus spreading equations to simulate the spread within a college community
- Orchestrated an in-depth literature review on disease spreading to ensure accurate techniques were being used
- Results were used to inform university policies and guidelines

Socio-Physics, Washington and Lee University, Undergraduate Researcher

Aug. 2019 – Dec. 2019

Prof. Irina Mazilu

Quantifying Conflict by Analyzing the Movement of Refugees

- Utilized Python to execute a mixture of rumor spreading and disease spreading models
- Presented data and correlations to peers and teachers, resulting in a lead discussion of policy and conflict

Additional Projects

Linguistic Representational Divergence in Large Language Models

Current

With Prof. Alexandra Siegel and Prof. Brian Keegan

With the large quality discrepancy in linguistic training data, we are assessing the representation of various Arabic historical information across LLMs, focusing on translation, content, and temporal variation within country-based identities portrayed through prompt-engineering

Wikipedia Affordance Perception Across Languages and Memories

Current

With Prof. Heather Ford, Anas Buhayh, Prof. Alexandra Siegel, Prof. Brian Keegan

There are different structural norms between language editions of Wikipedia. However, we don't know whether that influences Wikipedia affordance perception. Through surveys, retrospective interviews, and digital trace analysis, I investigate English and Arabic editors

*Wikipedia Article Ecosystem: Leveraging History, Navigation, Collaboration,
and Consumption Behaviors*

Current

Article similarity often focuses on content, however, there is a robust amount of underutilized data related to blue links, clickstreams, and shared editors that would contribute to their similarity. I utilize a multi-graph to leverage all of these data sources and identify an article's "ecosystem"

Extended Abstract: https://wikiworkshop.org/2025/paper/wikiworkshop_2025_paper_25.pdf

Remembering Afghanistan

Current

With Prof. Alexandra Siegel

Using English and Arabic based NLP tools, analyzing the misalignment of topic and sentiment in tweets in English and Arabic; focusing on the withdrawal of troops from Afghanistan and the year anniversary of the withdraw, proposing causal conclusions between the withdrawal of troops and geographic sentiment polarization

Multilingual Wikipedia Accessibility Dashboard

2023

With Prof. Ricarose Roque

Prioritizing user centered design foundations, creating a dashboard to leverage my multilingual data and

statistical analysis from the project with Wikimedia alongside LLM integration to encourage and facilitate cross-language content and context accessibility to multilingual and monolingual users

- Implemented qualitative and quantitative user research methods to identify user needs and preferences, resulting in the development of a dashboard that addressed diverse linguistic and cultural requirements. Conducted user-centric design evaluations to ensure the effectiveness and usability of the dashboard.

Publications and Presentations

Scholarly Publications

2025 Jones, H. Laurie et al. "Cross-Language Evolution of Divergent Collective Memory Around the Arab Spring." arXiv:2404.10706 **(under-review)**

Scholarly Presentations

2025 Jones, H. Laurie, Keegan, Brian C. (2024, January 11-11). *Divergent Collective Memory Processes Surrounding the Arab Spring* [MENA and Asia Political Methodology Conference]. NYU Abu Dhabi, Abu Dhabi, United Arab Emirates.

Jones, H. Laurie (2024, January 11-11). *Wikipedia Article Ecosystem: Leveraging History, Navigation, Collaboration, and Consumption Behaviors*. [Wiki Workshop 2025]. Virtual. <https://www.youtube.com/watch?v=lePKDgoxktc>

2024 Jones, H. Laurie, Keegan, Brian C. (2024, May 24-25). *Divergent Collective Memory Processes Surrounding the Arab Spring* [Visions in Methodology Conference]. Washington University in St. Louis, St. Louis, MO. United States

Jones, H. Laurie, Keegan, Brian C. (2024, June 20). *Temporal Clustering of Wikipedia Outlinks as Collective Memory Processes* [Wiki Workshop 2024]. Virtual. <https://www.youtube.com/watch?v=TrthZ-zJ8ow&feature=youtu.be>

2023 Jones, H. Laurie, Keegan, Brian C. (2023, June 8-9). "Network perspectives on collective memory processes across the Arab Spring in Arabic and English Wikipedias" [WikiHistories]. Virtual. <https://wikihistories.net/2023/04/13/laurie-jones-network-perspectives-on-collective-memory-processes-across-the-arab-spring-in-arabic-and-english-wikipedias/>

Invited Talks

2025 "What Do You Remember, an Introduction to Digitally Mediated Collective Memory". INFO 1101 Computation and Society, Information and Technology, University of Colorado Boulder

"Technology and the Recording of History". INFO 4601 Ethical and Policy Dimensions of Information and Technology, University of Colorado Boulder

2024 "The Ethics Behind Translation, Digital and Audible". INFO 4601 Ethical and Policy Dimensions of Information and Technology, University of Colorado Boulder

Teaching Experience

Graduate Part-Time Instructor, University of Colorado Boulder

Summer 2025 &

Summer 2024

INFO 2201: **Programming for Information Science 2:** Lecturing and grading 40 students advanced python as an introduction to Data Analysis and Data Science, focusing on research and industry-based application of web-scraping, Beautiful Soup, regex, numpy, and pandas

Teaching Assistant, University of Colorado Boulder

Spring 2025

INFO 1101: **Computation in Society:** Introducing non-quantitative students to the basic principles of software, programming, and AI as well as the fundamental role of technology in modern society

Fall 2025, Fall 2024

INFO 2201: **Programming for Information Science 2:** Led weekly recitations of 60 undergraduate students in advanced python including webscraping, Beautiful Soup, regex, numpy, and pandas

Spring 2023, Fall 2022,

& Fall 2021

INFO 1201: **Computational Reasoning:** Led weekly recitations of 50 undergraduate students for introductory Python and Javascript, created recitation activities and problem sets for homework assignments

Teaching Assistant, Washington and Lee University

Spring 2021

PHYSICS 265: **Modeling and Simulation of Physical Systems:** Created lesson plans and aided instruction of a class of 25 Physics Students. Worked to create physics related introductory - advanced coding assignments in python such as object-oriented programming, statistical analysis, and neural networks

Computer Science Peer Tutor, Washington and Lee University

Fall 2020, Spring 2020, Fall 2021, Spring 2021

- Tutored students in Python, Java, and C individually and in groups
- Specialized in Fundamentals of Programming I and II, Software Development, and Computer Organization

Leadership Experience

Graduate Student Mentor, University of Colorado Boulder

2022 - Current

- Guided and mentored multiple students transitioning from industry to graduate studies, providing tailored support and resources to facilitate a smooth academic and professional adjustment

President, Information Science Graduate Student Association

2023-2024

- Plans, coordinates, and conducts GSA meetings
- Communicated with administration or coordinate the delegation of that responsibility for specific initiatives
- Communicate GSA initiatives to the faculty's designated liaison to students
- Attends required external GSA-specific meetings (including financial, university, and as a student liaison for the Information Science department)
- Organizes and coordinates the GSA leadership team

NCAA Women's Varsity Golf Captain, Washington and Lee University

2017-2021

- Balance 25+ hours per week practice schedule with academics
- Able to make decisions and execute actions under pressure
- Lead team to 2 National Championships and two Conference Championship wins

Skills and Languages

Language

Arabic - moderately proficient; lived in Amman Jordan (summer 2019)

Coding Languages

- SQL: (proficient) database management, data retrieval and cleaning, data visualization
- Python: (expert) numpy, pandas, matplotlib, seaborn, NLTK, scikit-learn, networkx
- R: (expert) large scale data analysis and exploration, political & bioinformatic data science

Soft Skills

- Interdisciplinary Communication: communicated across computational disciplines (Physics, Computer Science, HCI, Math) to propose and present research, communicated across diverse fields (Middle Eastern Studies, Journalism, Political Science, Biology) to aid in research
- Project Management: led multiple teams of undergraduate students for research projects including work for government funded research projects.
- Communication: data-driven insights cross-department presentations in Los Alamos National Laboratory and International Conferences

Analytical Skills

- Technical Report Writing: Created written reports and presentations to effectively communicate experimental and computational projects
- Quantitative Modeling: statistical and machine learning methods for description, inference, classification, and visualization of quantitative, textual, and network data, A/B testing
- Information Retrieval: large-scale data extraction from APIs and databases, converting under-structured data into standard formats for analysis
- Research Design: interdisciplinary expertise in quantitative, computational, and mixed methodological designs for digital trace data analysis, git
- Regression Analysis: linear and logistic regression and norm analysis in python, taken advanced courses in R based Regression
- Data Visualization: multilingual visualization, matplotlib, Data Studio, Tableau